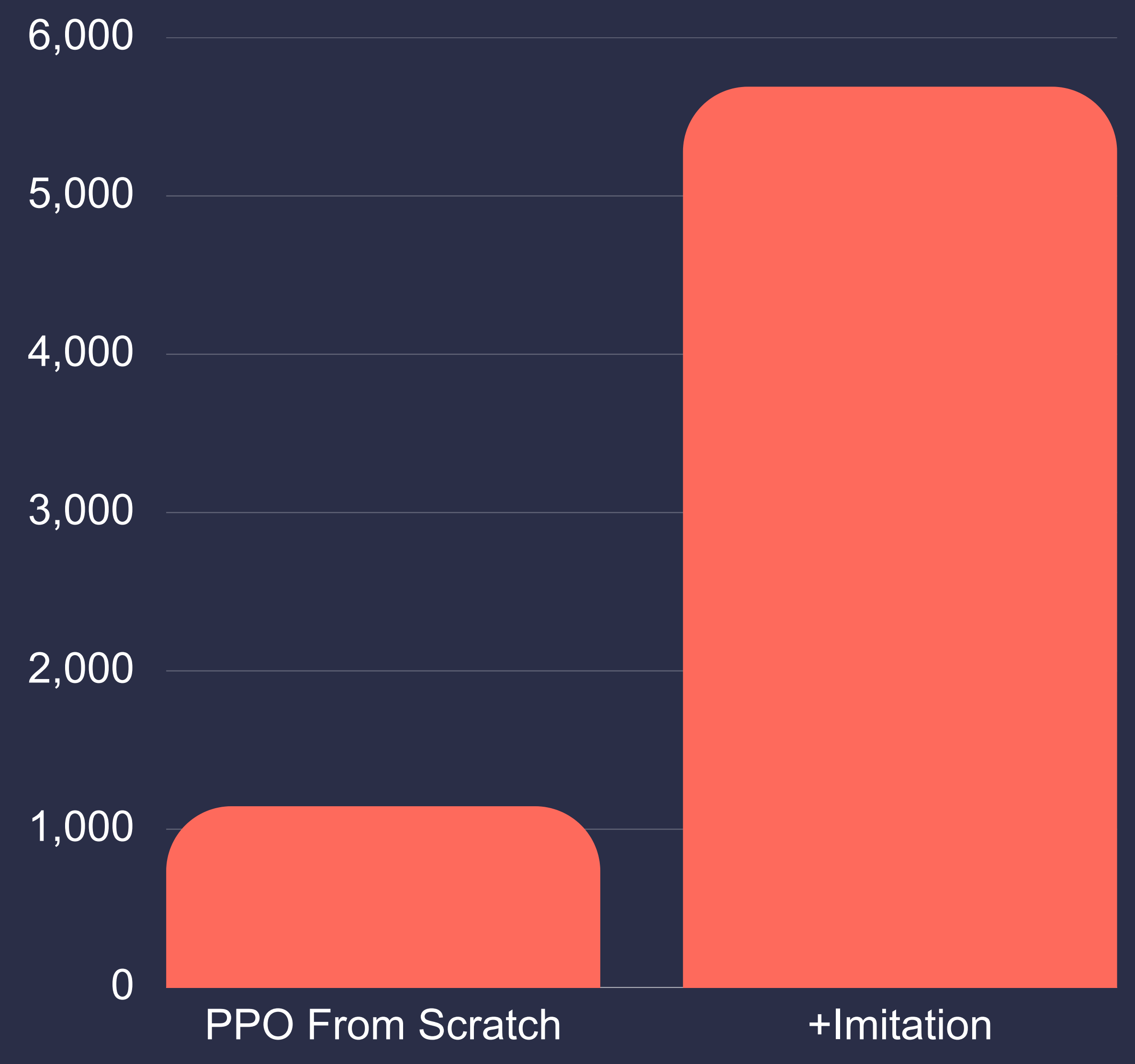
LEARNING TO MOVE

PRESS START

WALKING IS EXPENSIVE TO LEARN

- Get a body to walk from nothing, in a physics world.
- Brute force (PPO) works, but it is slow and costly.
- Our bet: train ONE master, then clone it cheaply.

Both agents trained for the SAME 1.5M steps



Walker2d eval return at 1.5M steps.
Imitation gets ~5x closer to the expert.

FIVE EXPERIMENTS

1. **EXPERT** — train a master ›
2. **DEMOS** — record it ›
3. **BC** — clone it ›
4. **DAGGER** — fix the clone ›
5. **WARM-START** — the payoff

BUILDING THE EXPERT

Train a master worth copying.
And the honest re-runs it took.

WHAT WE TRAIN,
AND WHAT "GOOD" MEANS

- PPO learns by trial and error: maximize forward motion, minus wasted effort.
- A shaky master makes shaky recordings, and that poisons every later stage.

WALKER2D

17 obs · 6 act · 2-D biped

BAR TO CLEAR

> 3000

ANT

27 obs · 8 act · 3-D, 4 legs

BAR TO CLEAR

> 4000

THE WALKER JOURNEY

Test, fail, retry. The variance was the problem, not the score.

2M steps

3010

UNSTABLE

❯

+ LR decay · 4M

4616

TIGHTER

❯

+ VecNormalize · 8M

6043

STONGER

Decaying the learning rate and normalizing observations (VecNormalize) steadied it.
VecNormalize was the single decisive lever.

THE ANT JOURNEY, AND ITS LESSONS

Walker recipe

2422

PLATEAU

›

+ VecNormalize

68%

GATE FAILS

›

tuned · 10M

6293

GATE OK

RECIPES DO NOT TRANSFER

The tuned recipe made Walker worse: 1640,
below threshold.

DO NOT JUDGE A NOISY MIDDLE

We killed Ant at 5M thinking it failed.
The full run was fine.

RECORD THE MASTER

100 expert episodes, deterministic actions, checked against a quality gate (≥ 90% above ⅔ of the mean).

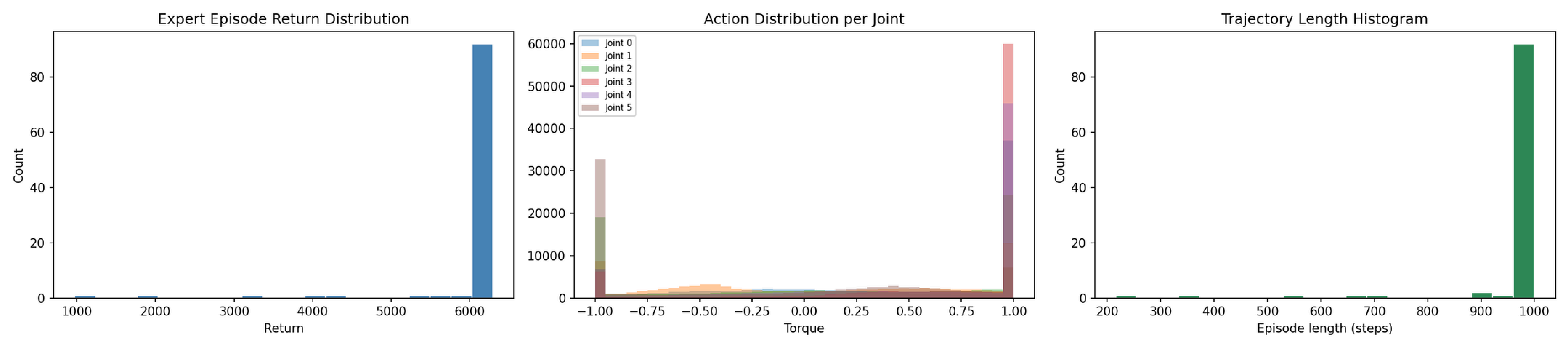
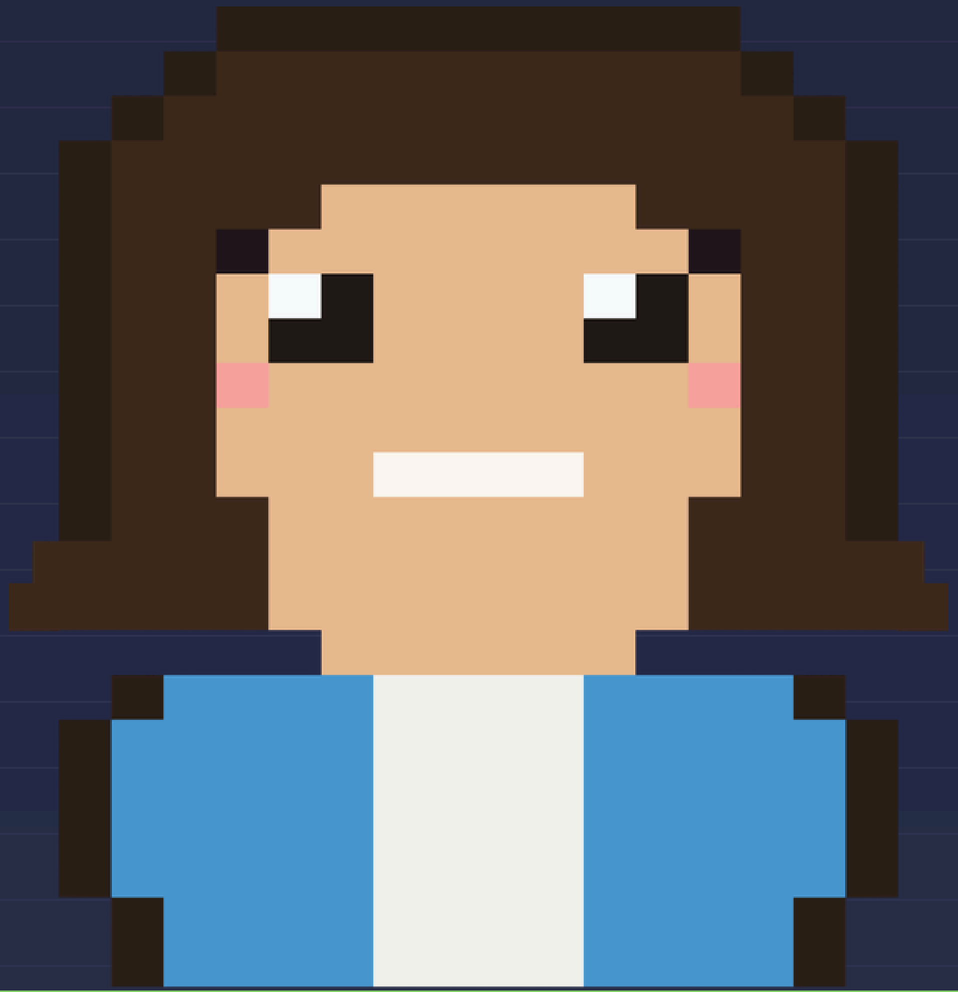


Figure 2 — Walker2d demonstrations: return distribution, per-joint actions (saturated near ±1), trajectory lengths near the 1000-step cap.



BEHAVIOURAL CLONING

Can a student learn to walk
by pure imitation?

THE STUDENT’S JOB

- Behavioural Cloning treats imitation as supervised regression.
- Predict the expert’s action from the state. No reward, no environment.
- Cheap and appealing, but it never sees its own mistakes.

STATE › STUDENT MLP › ACTION

$$\text{minimize} \;\; \| \text{student(state)} - \text{expert action} \|^2$$

TWO BUGS, TWO FIXES

raw observations

1246

UNDERTRAINED

›

normalize inputs

jumps

FIXED

50 epochs

47%

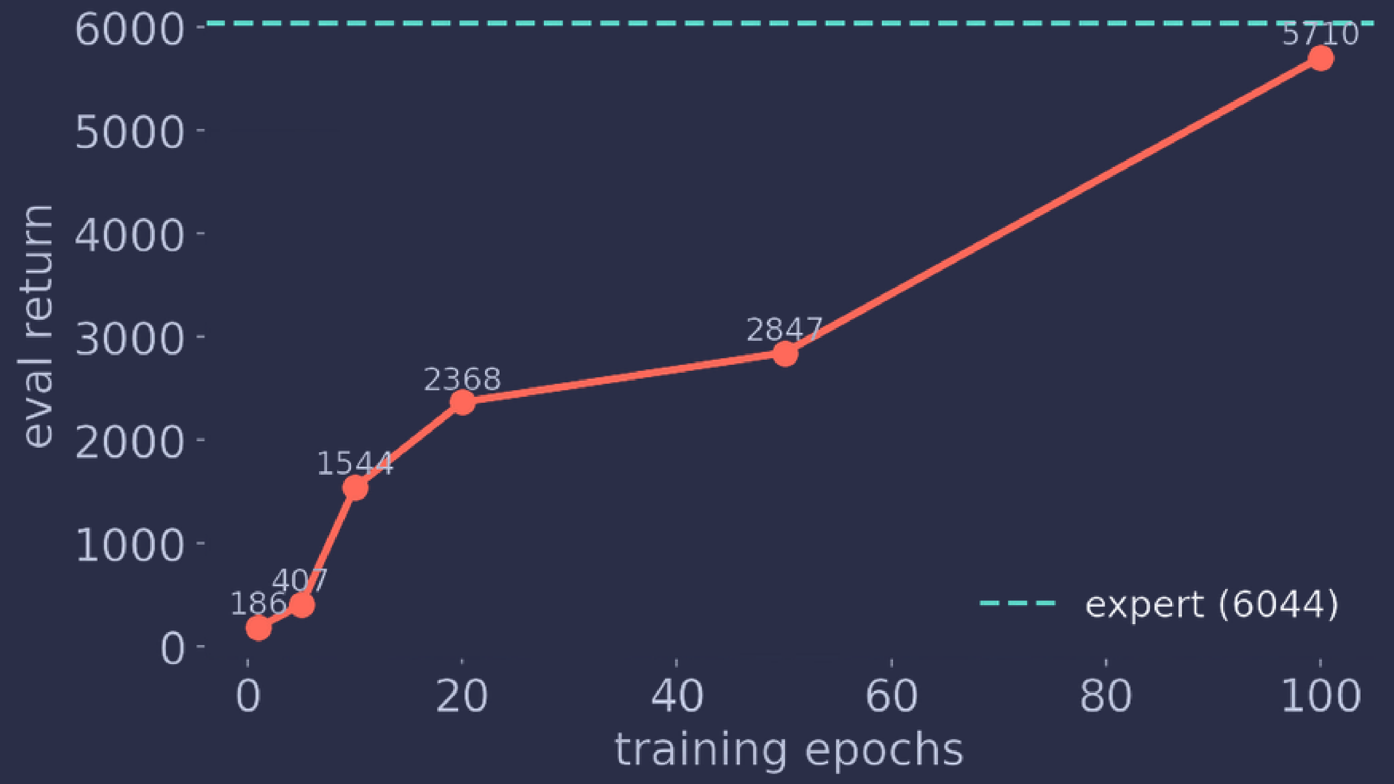
UNDERTRAINED

›

early stopping

95%

FIXED



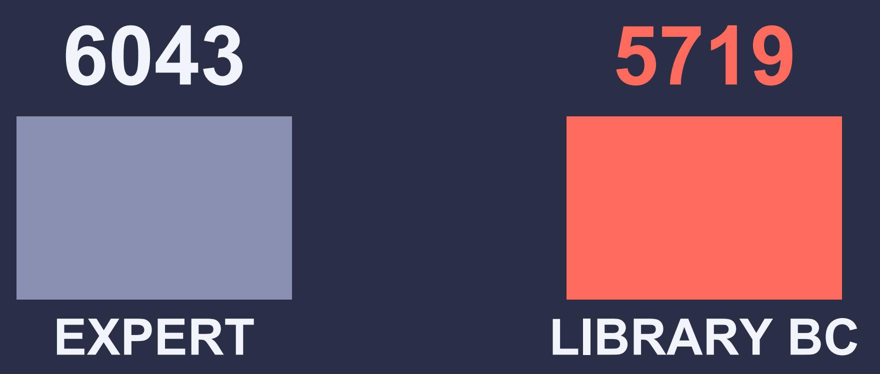
Epoch sweep: the 47% was undertraining, not a ceiling.

HOW CLOSE CAN PURE COPYING GET?

Library BC matches the expert; a from-scratch PyTorch BC validates our code.

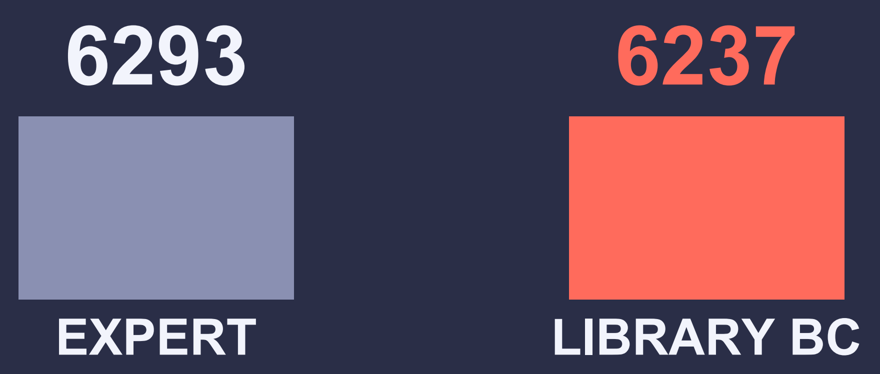
WALKER2D

95%

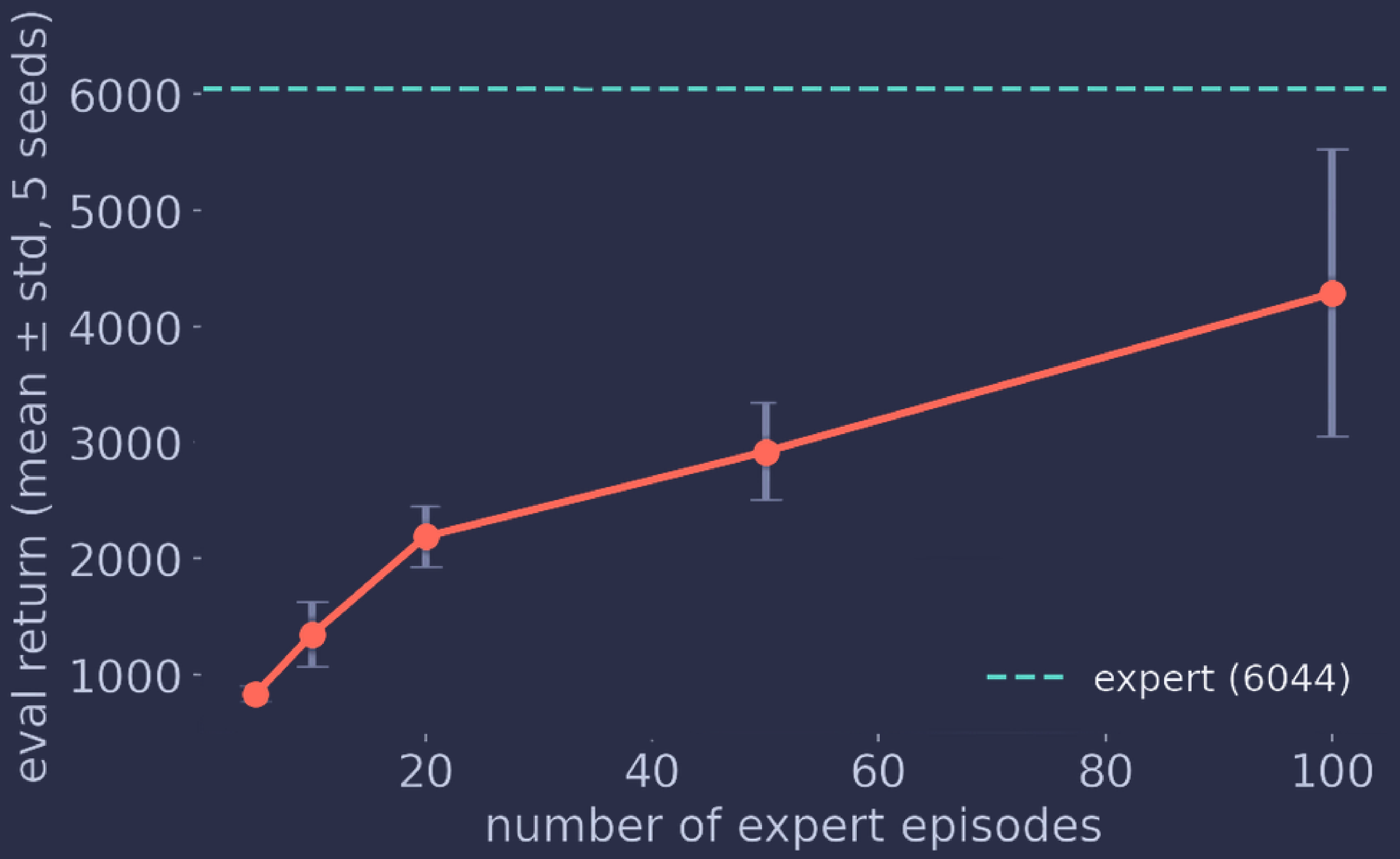


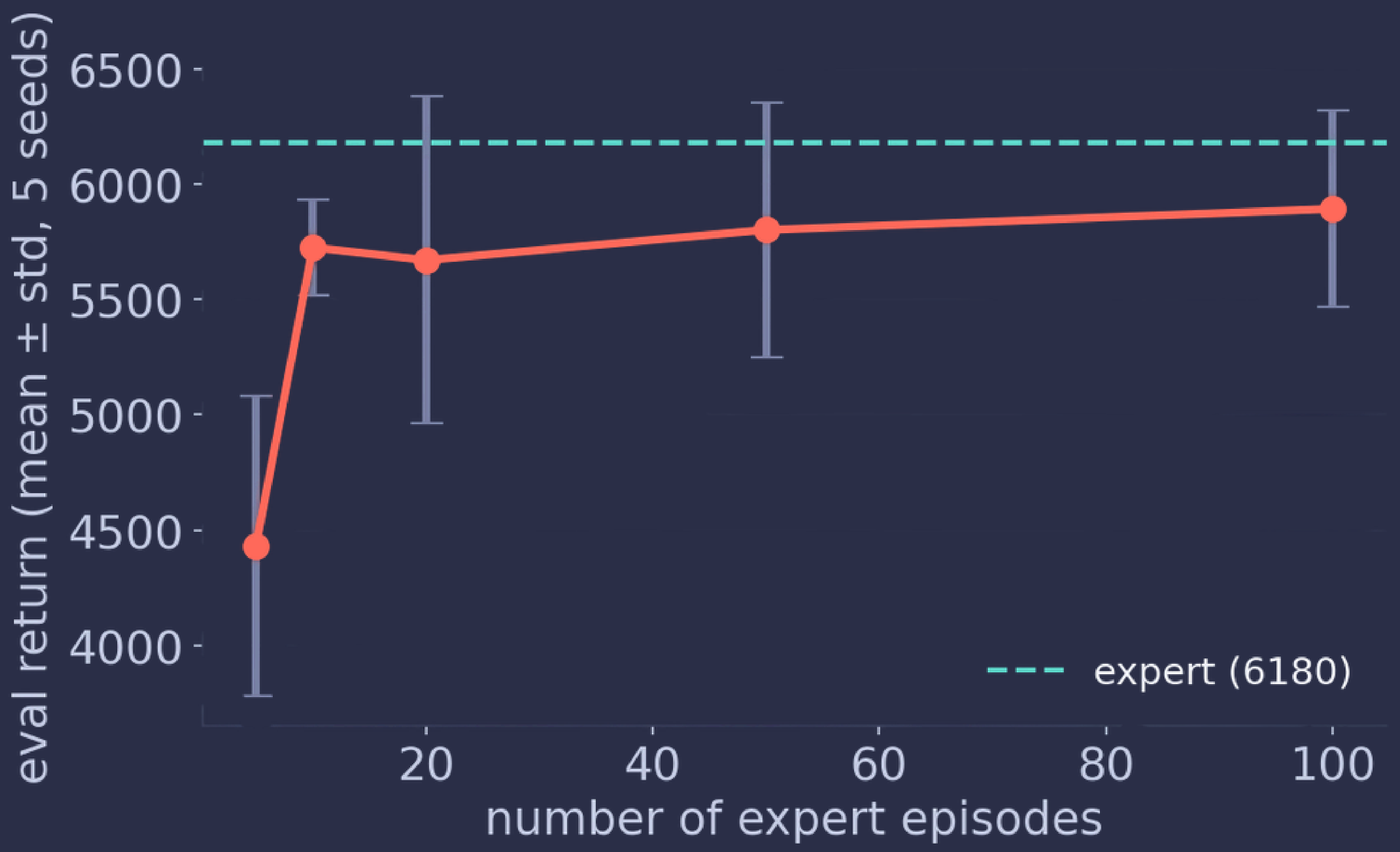
ANT

99%



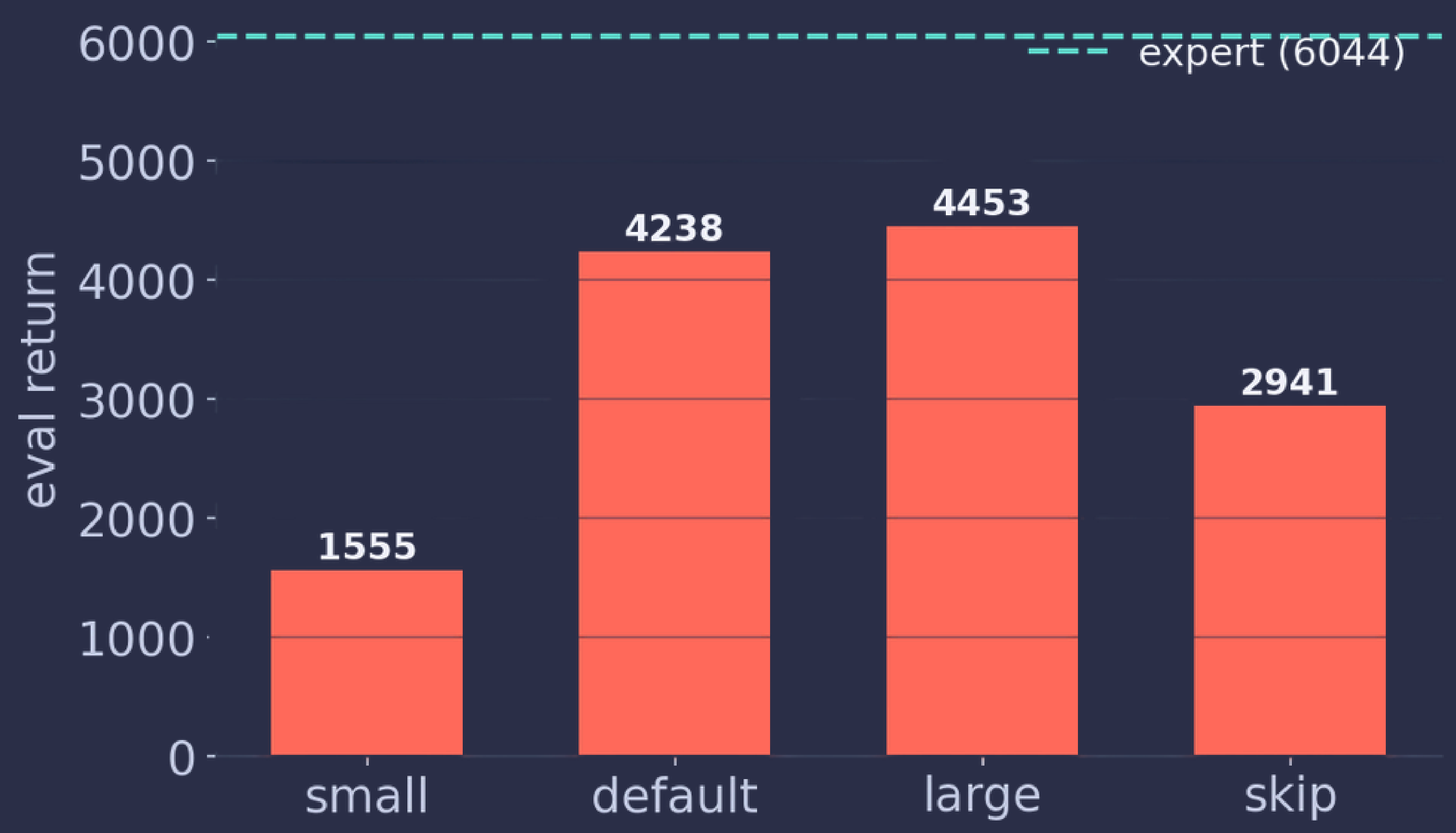
HOW MUCH EXPERT DATA IS ENOUGH?

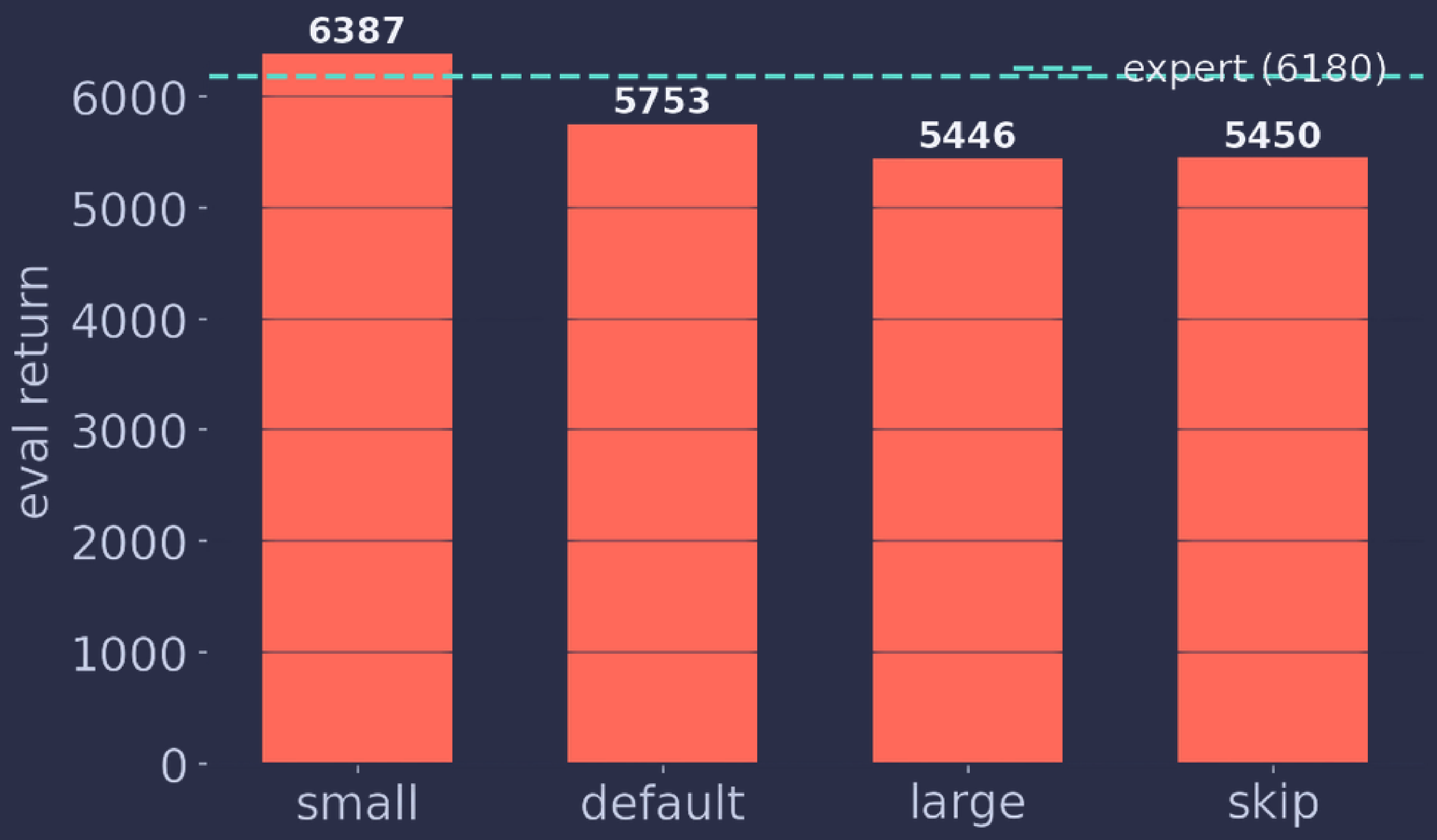




Climbs fast to ~50 episodes, then flattens. 5–10 episodes are low and high-variance.
~50 demos capture most of the achievable performance.

DOES THE STUDENT'S NETWORK MATTER?





Environment-dependent:

Walker needs a bigger net (512,512 wins, 64,64 underfits);
Ant fits easily and barely cares.
Capacity matters on the harder env.

A LOW LOSS IS NOT A GOOD WALKER

- Online return and offline action-MSE do not track each other.
- At a fixed low validation MSE, evaluation return still varies by 1000+ across seeds.
- Imitation is judged sequentially, so small per-step errors compound.

THE TAKEAWAY

A low supervised loss is necessary but not sufficient.
Always evaluate the student in the environment, not just on the dataset.

DAGGER:
FIXING THE FLAW

BC’s hidden weakness, and the fix that reaches expert level.

WHY PURE COPYING DRIFTS

- BC only ever saw the expert’s path.
- Off that path it has no idea and small errors snowball.
- Great on short tasks; fails on long-horizon locomotion.

errors compound: $O(\varepsilon T^2)$ — quadratic in the horizon length

This is the central weakness of plain BC. DAgger is built to fix exactly this.

DAGGER: A LIVE TUTOR

4 epochs / iter

2402

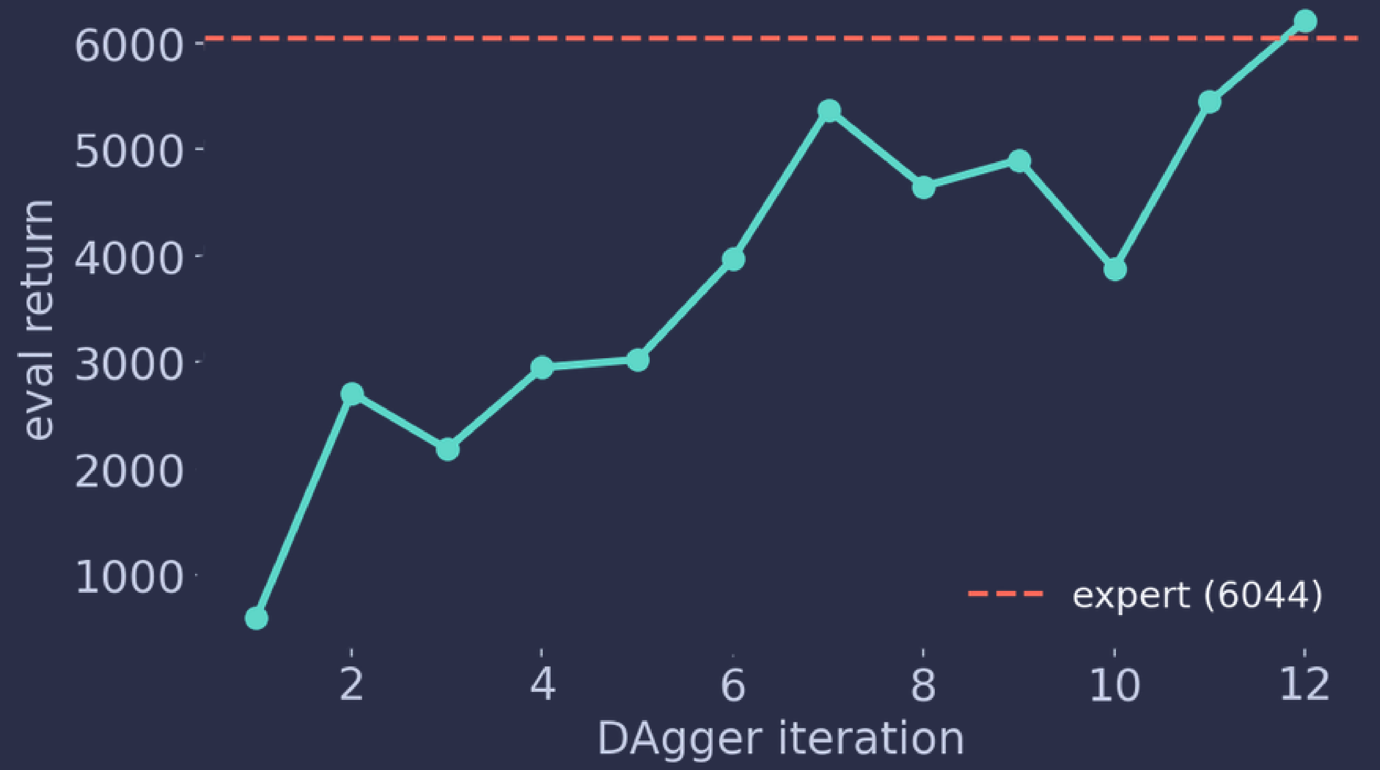
Looks Worse

›

fair: 12 iters

6208

Beats BC

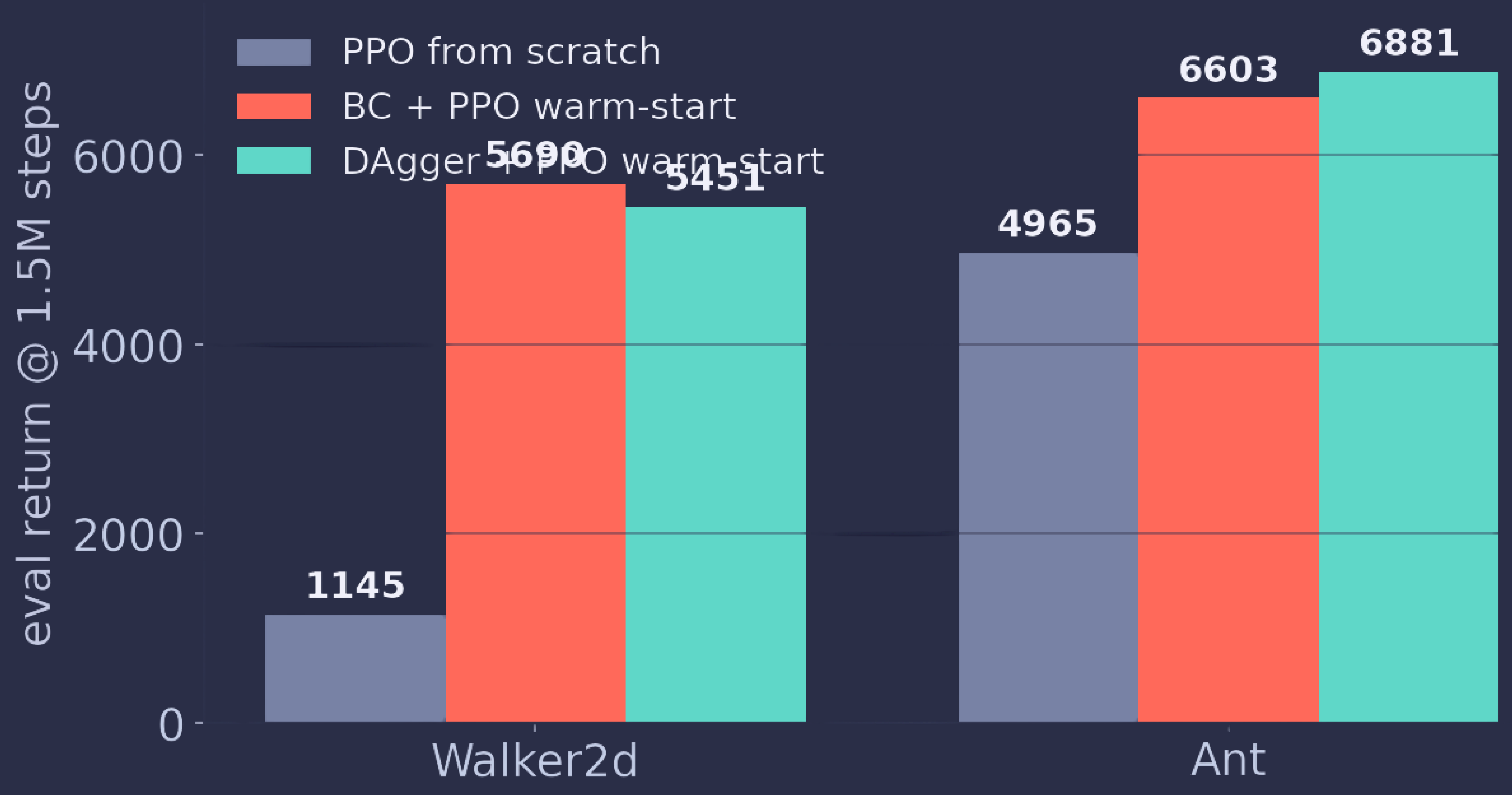


The first run was unfair, far less training than BC. With a fair budget, DAgger matches or beats BC everywhere and climbs ~600 → 5562 as the expert labels recovery behaviour.

THE PAYOFF

Does imitation actually make PPO cheaper?

IMITATION AS A HEAD START

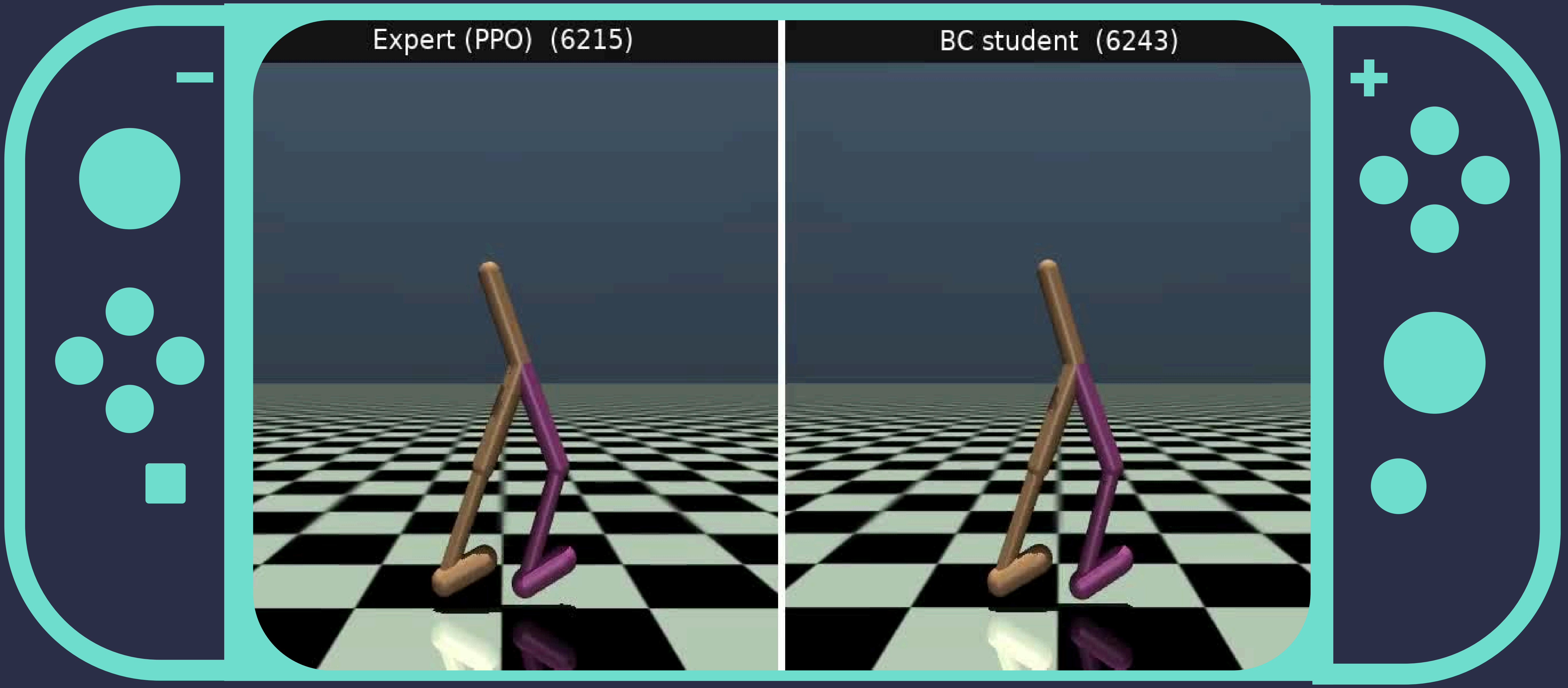


~5x FEWER STEPS

At 1.5M steps, warm-started PPO is near expert while from-scratch PPO is still flailing. Largest gains where learning from scratch is hardest.

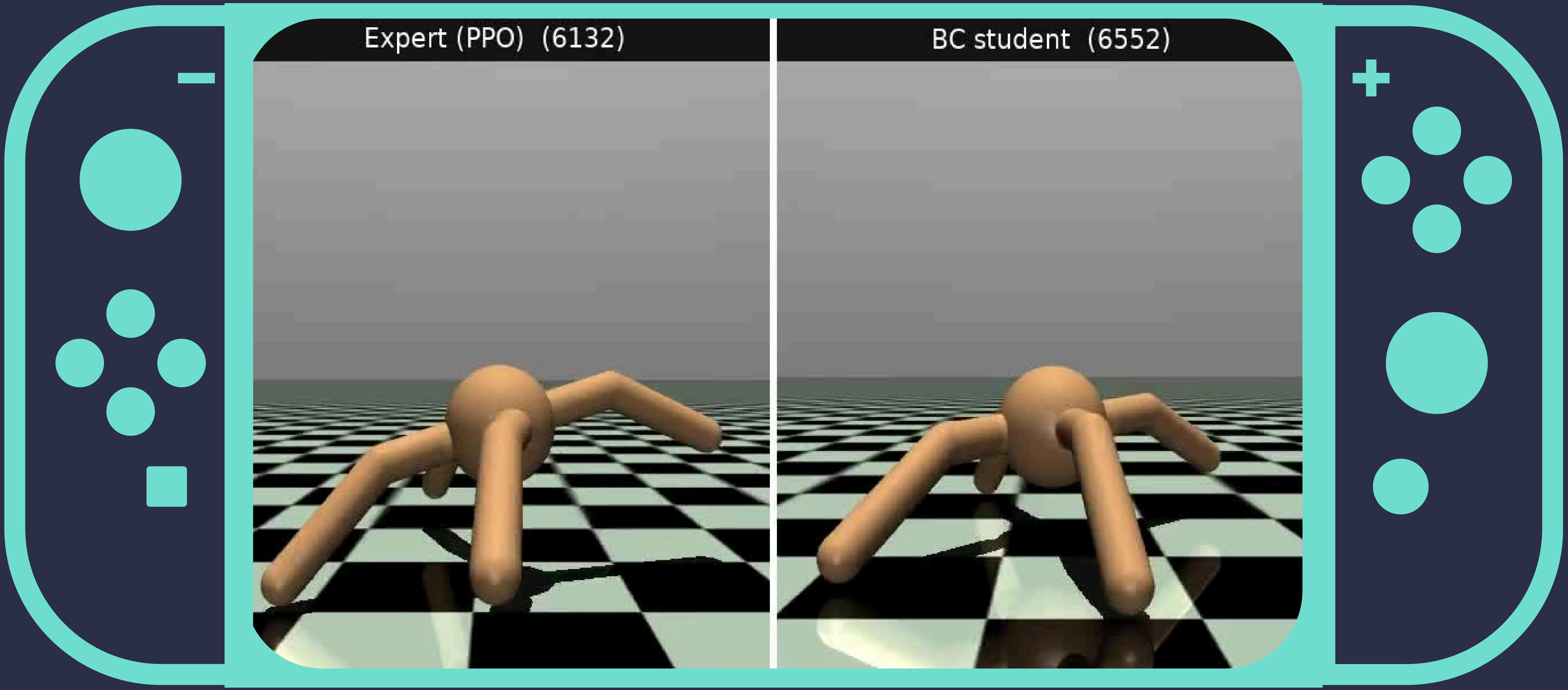
Walker dips early under its larger LR then recovers; Ant holds from step 0.

SEE IT MOVE



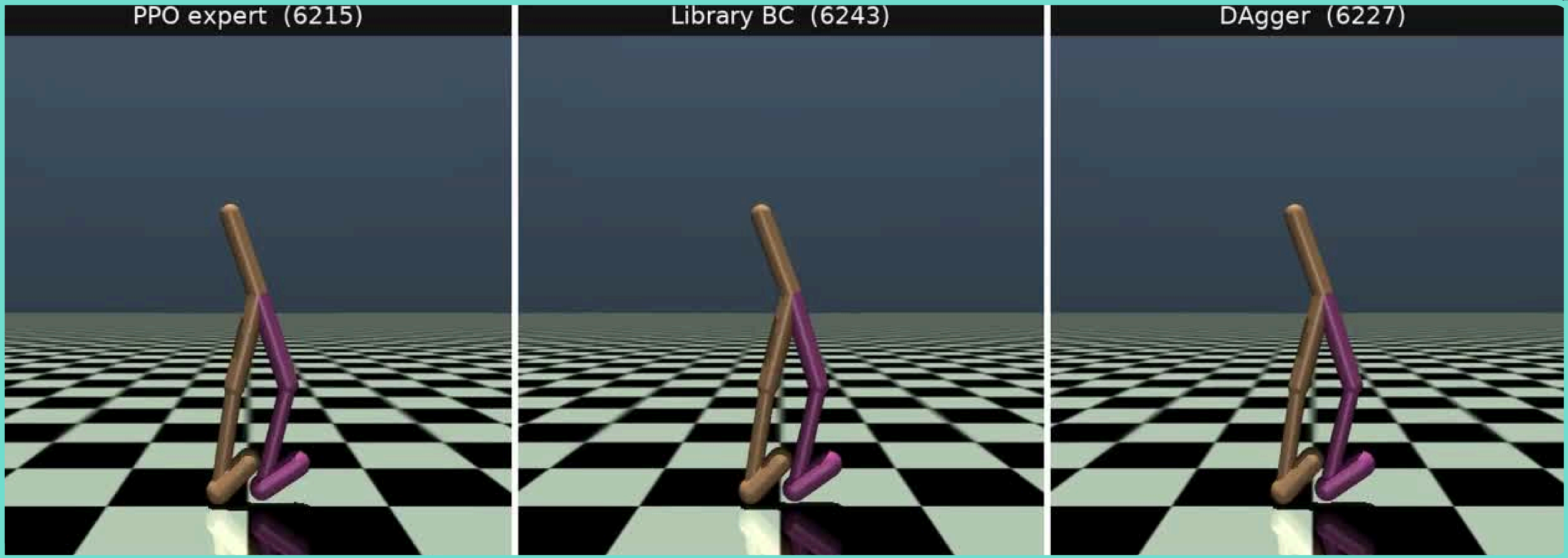
WALKER2D · EXPERT vs STUDENT

SEE IT MOVE

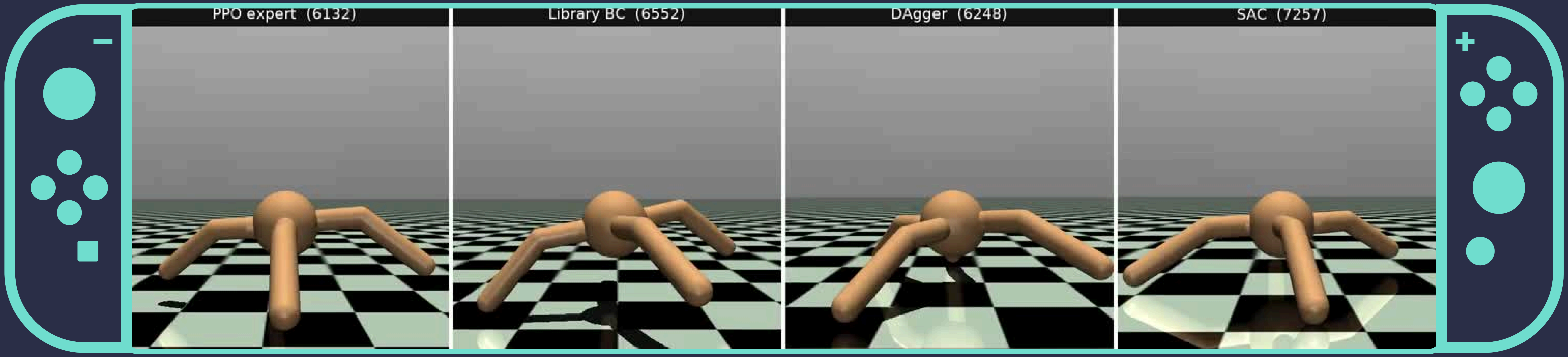


ANT · EXPERT vs STUDENT

SEE IT MOVE



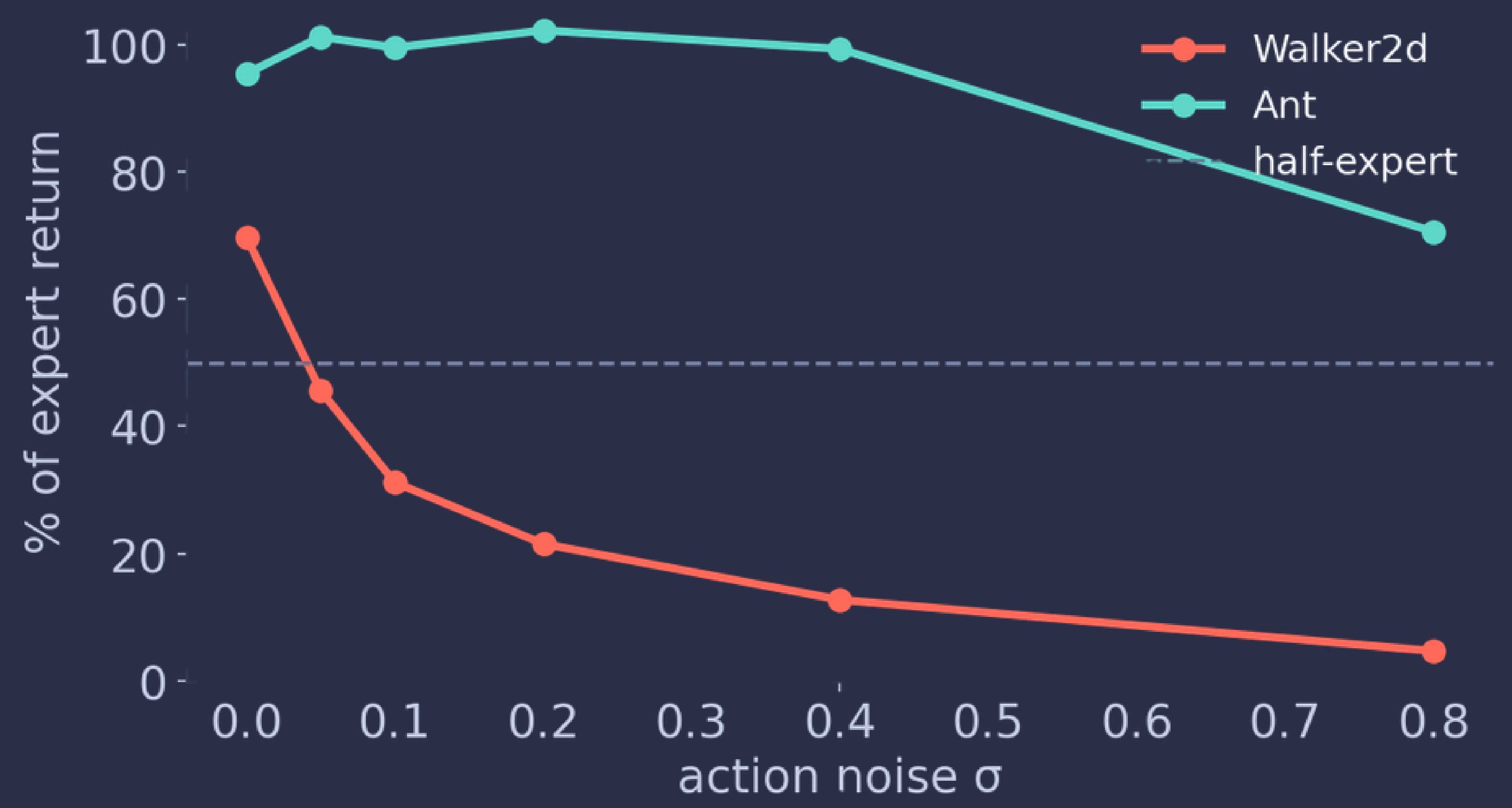
SEE IT MOVE

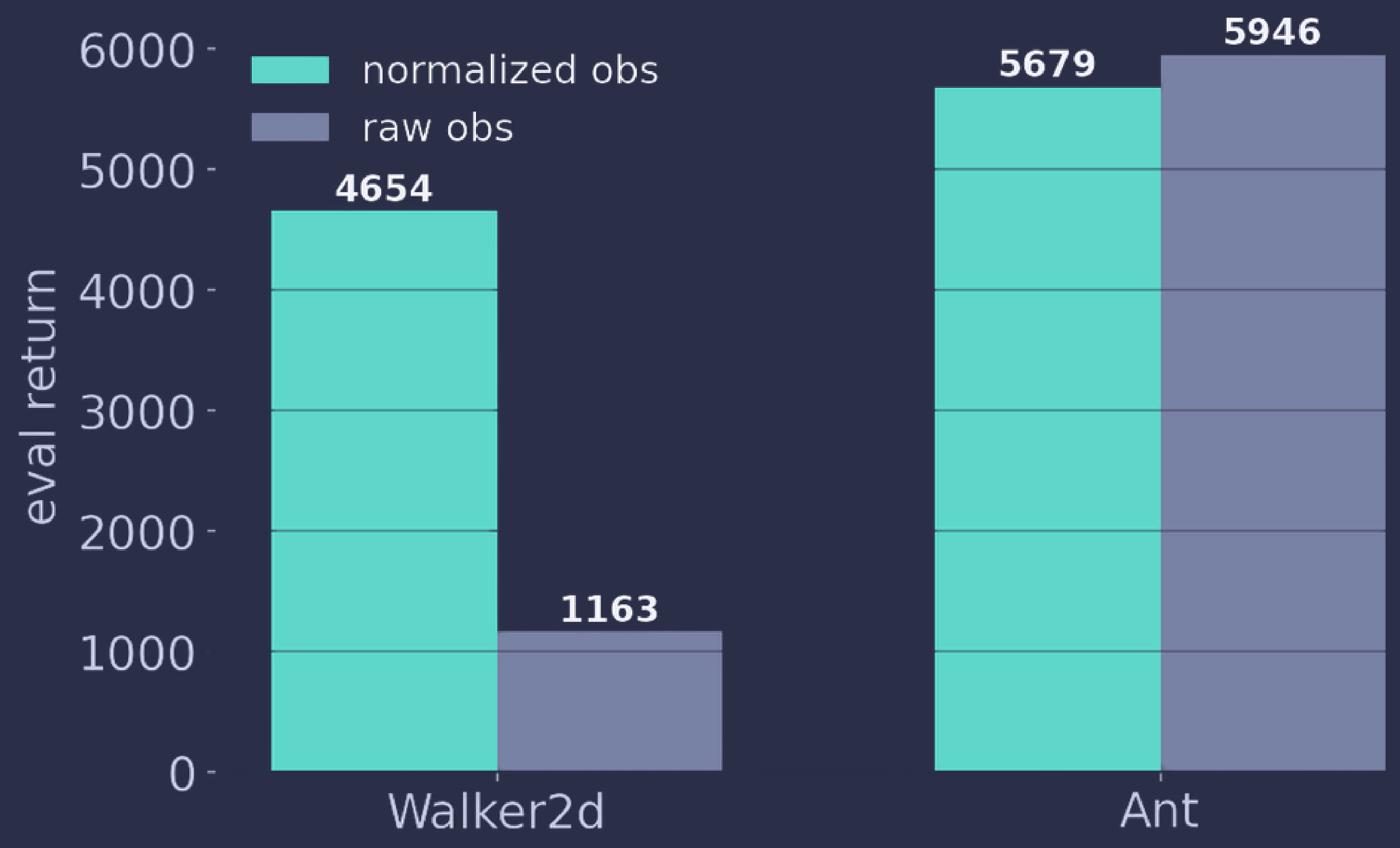


STRESS-TESTS & MEANING

We pushed it until it broke and broke it some more.

WE PUSHED IT UNTIL IT BROKE





Noisy expert: Walker collapses at σ=0.05, Ant holds through 0.4.
Normalization: decisive for Walker (4x), irrelevant for Ant.
Same theme: Walker is fragile, Ant is robust.

A RIVAL, AND THE PATTERN

SAC (off-policy)

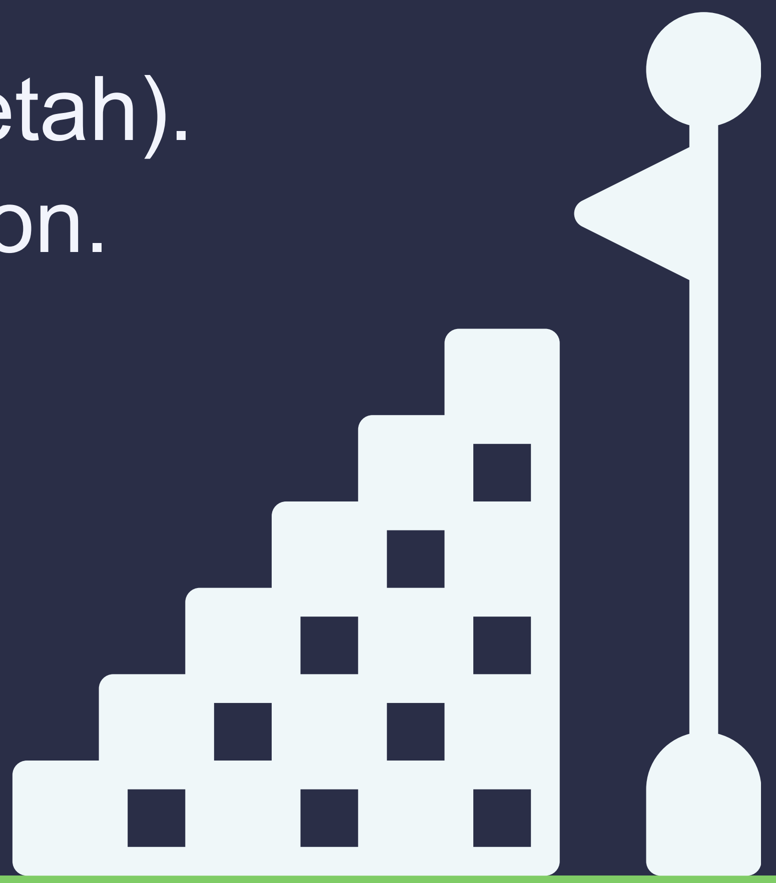
- Beats our PPO Ant expert: 7295 vs 6293.
- Uses ~3x less practice (3M vs 10M steps).
- On HalfCheetah it clears 15000.

WALKER2D vs ANT (RQ6)

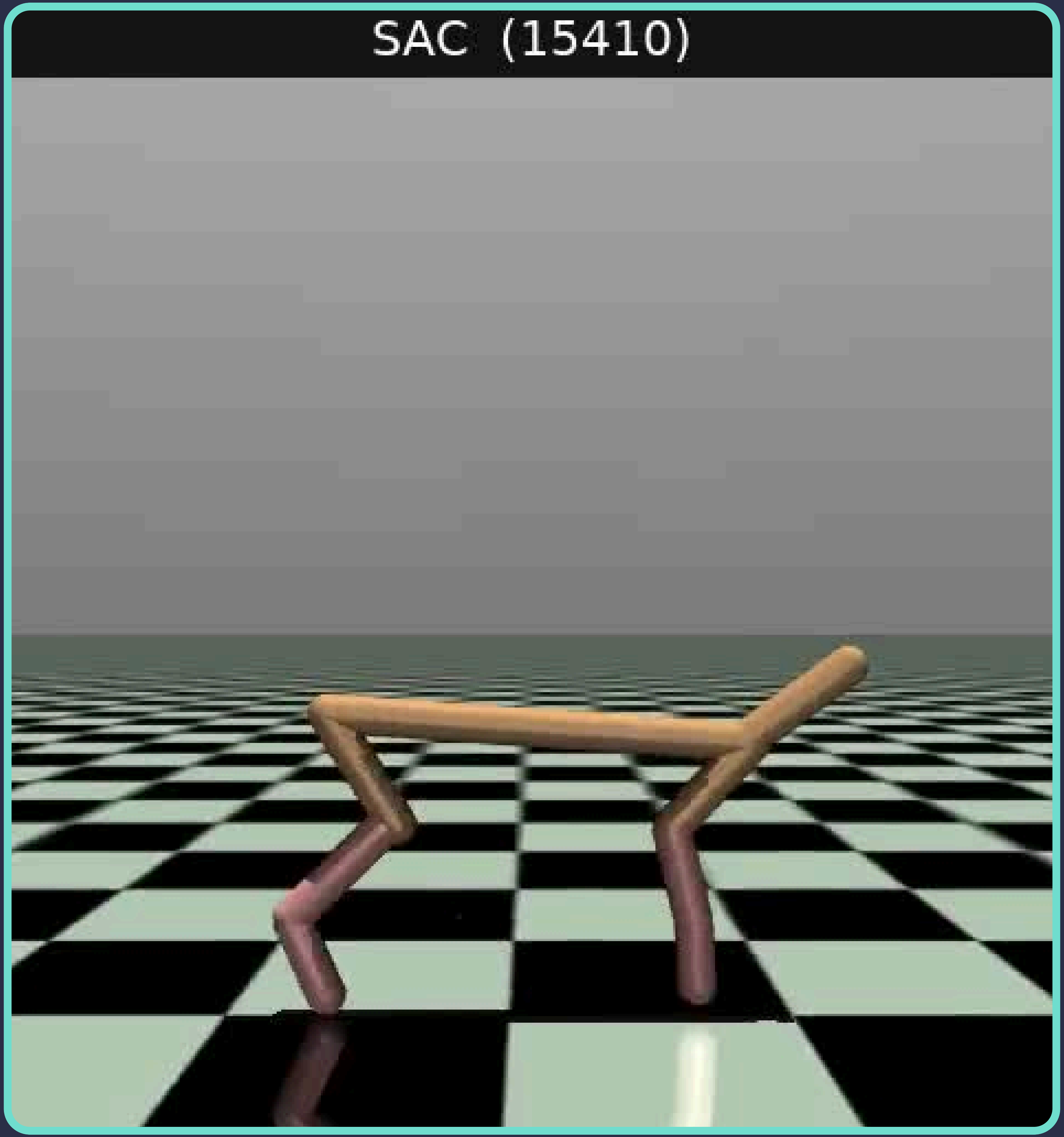
Ant is bigger and 3-D, yet EASIER to imitate (99% vs 95%, architecture barely matters, robust to noise). Training difficulty and imitation difficulty are different axes.

THE EXTRA MILE

- Both Walker2d AND Ant, end to end (the brief allowed one).
- An off-policy algorithm (SAC) and an off-brief env (HalfCheetah).
- A second Walker expert (4627) for a both-experts comparison.
- E1 + E2 bonus studies, 5 seeds each.
- Multi-policy comparison videos (PPO / BC / DAgger / SAC).
- A fully reproducible, seeded pipeline plus a Colab runner.



SEE IT MOVE



WHAT SURPRISED US

- The tuned recipe made Walker WORSE, recipes do not transfer.
- A low BC loss did not mean a good policy.
- SAC beat our PPO expert with far less practice.
- Killing a run from its noisy middle led us to a wrong conclusion.

THE SIX QUESTIONS, ANSWERED

Central question: yes, imitation cuts PPO’s sample complexity, decisively.

RQ1 BC reaches 95–99%; the limit was training budget, not a ceiling.

RQ2 Performance saturates around 50 demonstrations.

RQ3 A low loss does not imply a good policy (errors compound).

RQ4 Architecture matters on the harder env (Walker), not Ant.

RQ5 DAgger fixes covariate shift and reaches expert level.

RQ6 Ant is easier to imitate than Walker despite higher dimensionality.

GAME OVER